

AE 2610, Lab #1

Tensile Testing

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Lab Group A02-X

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Introduction

In this lab, numerous characteristics of metals needed to be measured. First, all the reduced section dimensions were required for each specimen. This included measuring the metal's final and initial lengths, the width and thickness of the reduced section before testing as well as the width and thickness of both the uniformly deformed and fractured regions after failure. Second, the initial gauge lengths of the extensometer and the two raw csv data files generated during testing needed to be considered. To ensure accuracy and safety, this lab was conducted in a controlled environment under the guidance of a teaching assistant. Data was gathered utilizing an Instron 5982 load frame to deliver axial force, a pre-wired strain gauge to measure material strain, and an extensometer to record the relative displacement of the reduced section over time. This data was gathered to determine other material properties like ultimate tensile stress, fracture strain, and strength coefficient. Ultimately, this data enabled the discovery of the true identity of mystery specimen.

Data Results

Raw Data

It is important to note that before any measurements were made, the gauge length, L_0 , for the extensometer was set to 2 inches for both specimens. Additionally, the raw csv data file for Al 2024-T3 is summarized in Figure 1 and the raw csv data file for the mystery material is summarized in Figure 2.

Prior to Testing: Aluminum

- Length of Reduced Section: 70 mm
- Thickness: 2.02 mm
- Width: 13 mm

Prior to Testing: Mystery Metal

- Length of Reduced Section: 69 mm
- Thickness: 2.29 mm
- Width: 13 mm

Following Failure: Aluminum

- Length of Reduced: 80 mm
- Width of Fracture Region: 11.7 mm
- Thickness of Fracture Region: 1.87 mm
- Width of Uniform Deformation Region: 11.95 mm
- Thickness of Uniform Deformation Region: 1.87 mm

Following Failure: Mystery Metal

- Length of Reduced: 91 mm
- Width of Fracture Region: 9.33 mm
- Thickness of Fracture Region: 1.53 mm
- Width of Uniform Deformation Region: 10.69 mm
- Thickness of Uniform Deformation Region: 1.65 mm

Reduced Data

The data reduction section of this lab can be broken down into four major parts:

Compute Engineering Stress

To begin, the first objective in this lab was to compute the engineering strain using the force calculated by the load cell and the initial cross-sectional area of the reduced section of each specimen. Cross-sectional area was computed by multiplying the reduced section width and thickness found in Table I, and engineering stress was computed using the following equation:

$$\sigma = \frac{P}{A_0} \quad (1)$$

* σ – Stress, P – Force, A_0 – Initial Cross Sectional Area

Utilizing equation (1) in Excel, the function was applied to every data point and graphed. The engineering stress vs. strain for Al 2024-T3 is represented in Figure 3 and that of the mystery material is represented in Figure 4.

Utilize Linear Elastic Region to Estimate Young's Modulus

The next requirement was to estimate the Young's Modulus for each material using the linear elastic region in the aforementioned stress vs. strain figures. This was done by first isolating the initial linear section because it enabled the use of the equation:

$$\sigma = E\varepsilon \quad (2)$$

* E – Young's Modulus

Once the elastic section was isolated, the Young's Modulus was calculated using a linear approximation function in Excel because, when rewritten, equation (2) shows that the Young's Modulus is simply the slope of stress-strain curve. The final values for the each material's Young's Modulus is found in Table II.

Utilize Load and Strain Data to Estimate Material Characteristics

In this point of the lab, there were many material characteristics that were able to be calculated. These characteristics were found through a series of equations included below, and all results are found in Table III (Al 2024-3) and Table IV (Mystery Material).

- Engineering Ultimate Tensile Stress:

$$UTS_{eng} = \frac{P_{max}}{A_0} \quad (3)$$

* P_{max} – Maximum Applied Force

- True Ultimate Tensile Stress:

$$UTS_{true} = \sigma_{max}(1 + \varepsilon_{max}) \quad (4)$$

* σ_{max} – Stress at Maximum Applied Force, ε_{max} – Strain at Maximum Applied Force

- True Fracture Stress:

$$\sigma_{fract,true} = \frac{P_f}{A_f} \quad (5)$$

* P_f - Applied Load at Fracture, A_f – Cross-Sectional Area at Fracture

- Engineering Fracture Stress:

$$\sigma_{fract,eng} = \frac{P_f}{A_0} \quad (6)$$

- True Ultimate Tensile Strain (Extensometer):

$$UTS_{true,ext} = \ln(1 + \varepsilon_{max}) \quad (7)$$

- Engineering Fracture Strain (Dimensions):

$$\varepsilon_{fract,eng} = \frac{\delta}{L_0} = \frac{L_f - L_0}{L_0} \quad (8)$$

* δ – Elongation, L_0 - Initial Reduced Section Length, L_f – Reduced Section Length after Failure

- True Fracture Strain (Dimensions):

$$\varepsilon_{fract,true} = \ln\left(\frac{L_f}{L_0}\right) = \ln\left(\frac{A_0}{A_f}\right) \quad (9)$$

* A_f – Area of Fracture Region

- True Fracture Strain (Extensometer):

$$\varepsilon_{fract,true,ext} = \ln(1 + \varepsilon_f) \quad (10)$$

* ε_f – Strain at Failure

Determine Strain Hardening Index and Strength Coefficient

Finally, this lab required the calculation of both strain hardening index n and strength coefficient K using data from the plastic region of the stress-strain curves. In the plastic region, ductile metals satisfy the following equation:

$$\sigma_T = K \varepsilon_T^n \quad (11)$$

Taking the natural log on both sides of equation (11) results in:

$$\ln(\sigma_T) = \ln(K) + n \ln(\varepsilon_T) \quad (12)$$

When equation (12) is plotted, the strain hardening index becomes the slope and the natural log of the strength coefficient becomes the y-intercept when using a linear approximation.

Information regarding the specific strain hardening indexes and strength coefficients of the specimens can be found in Table V.

Discussion

Additional Discussion

- This lab was conducted in a safe and controlled environment with each member aware of the procedures and a protection glass separating the group from the load frame.
- While conducting the lab and subsequent data reduction, it is important to emphasize that the y-intercept of the linearized strain hardening law equation is not K but the natural log of K.
- Information regarding the actual Young's Modulus for Al 2024-T3 was found in an academic report by Aditya Prabowo, Ridwan Ridwan, and Teguh Muttaqie.
- Even in a relatively small sample of metal, the deviation of true stress vs engineering stress is still significant as seen on Figures 5 and 6.
- The final identity of the mystery metal was able to be narrowed down by comparing many of the material characteristics with those of the metals listed in the pre-lab.

(Utilizing information from official sources)

Tables and Figures

Table I. Initial and final physical measurements of the Al 2024-T3 and mystery specimens.

Material	Al 2024-T3	Mystery
Reduced Section Length	70.0 mm	69.0 mm
Reduced Section Width	13.0 mm	12.6 mm
Reduced Section Thickness	2.02 mm	2.29 mm
Fracture Region Width	11.7 mm	9.33 mm
Fracture Region Thickness	1.87 mm	1.53 mm
Uniform Deformation Region Width	11.95 mm	10.69 mm
Uniform Deformation Region Thickness	1.87 mm	1.65 mm
Final Length of Reduced Section	80.0 mm	91.0 mm
Guage Length of Extensometer	2 in	2 in

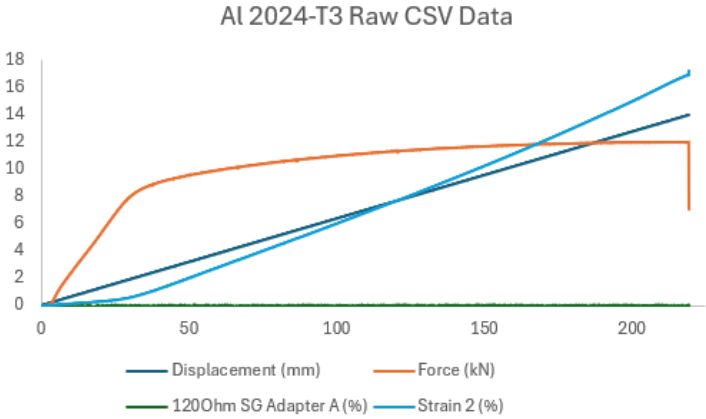


Figure 1. Al 2024-T3 Raw CSV Data

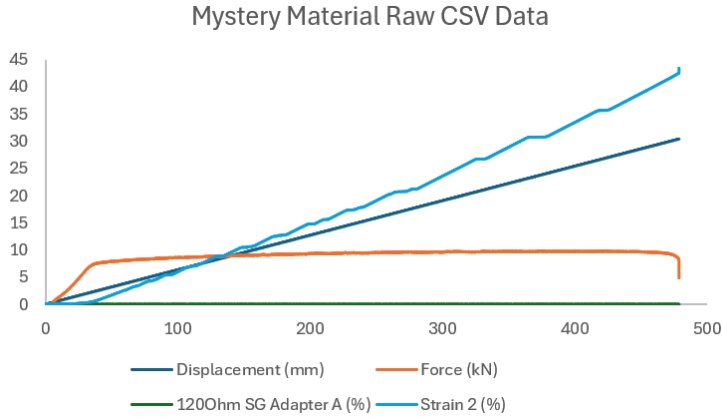


Figure 2. Mystery Material Raw CSV Data

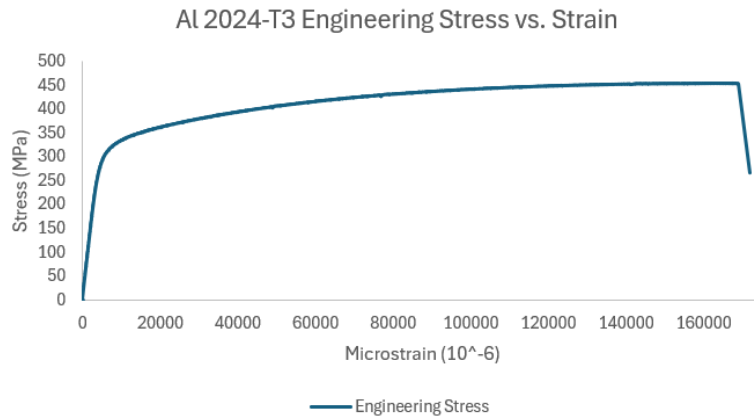


Figure 3. Engineering stress vs. strain curve for Al 2024-T3

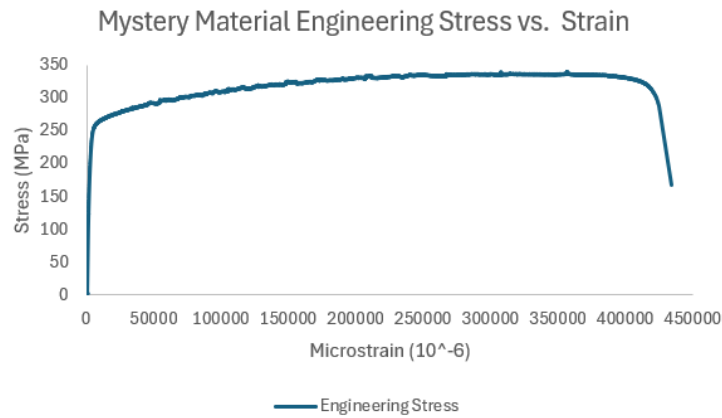


Figure 4. Engineering stress vs strain curve for the mystery specimen.

Table II. Young's modulus analysis for Al 2024-T3 and mystery specimen.

Material	Al 2024-T3 (Estimated)	Mystery	Al 2024-T3 (Actual)
Young's Modulus	71.96 GPa	103.74 GPa	73.4 GPa

Table III. Characteristic properties of Al 2024-T3

Al 2024-T3	ksi	MPa	Unitless
Engineering Ultimate Tensile Stress	65.94	454.64	X

True Ultimate Tensile Stress	76.67	528.66	X
Engineering Fracture Stress	38.71	266.91	X
True Fracture Stress	46.46	320.35	X
True Ultimate Tensile Strain (Extensometer)	X	X	0.151
True Fracture Strain (Extensometer)	X	X	0.159
True Fracture Strain (Dimension)	X	X	0.183
Engineering Fracture Strain (Dimensions)	X	X	0.143

Table IV. Characteristic properties of mystery specimen.

Mystery Specimen	ksi	MPa	Unitless
Engineering Ultimate Tensile Stress	49.35	340.29	X
True Ultimate Tensile Stress	66.95	461.60	X
Engineering Fracture Stress	24.27	167.34	X
True Fracture Stress	49.10	338.37	X
True Ultimate Tensile Strain (Extensometer)	X	X	0.305

True Fracture Strain (Extensometer)	X	X	0.361
True Fracture Strain (Dimension)	X	X	0.704
Engineering Fracture Strain (Dimensions)	X	X	0.319

Table V. Strain hardening index and strength coefficient of both specimens.

Material	Strain Hardening Index (n)	Strength Coefficient (k)
Al 2024-T3	0.1108	619.55 GPa
Mystery	0.1267	489.90 GPa

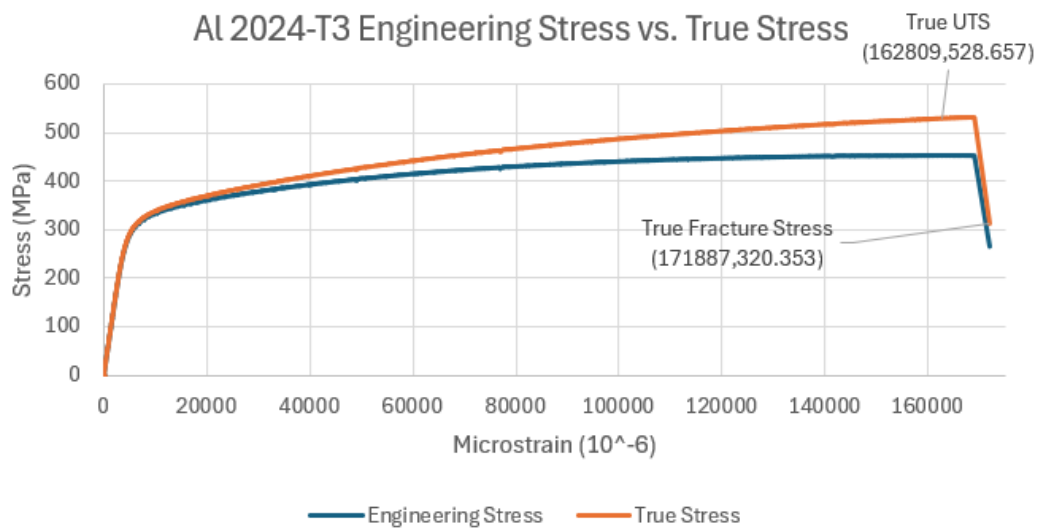


Figure 5. Engineering stress vs. true stress curves for Al 2024-T3.

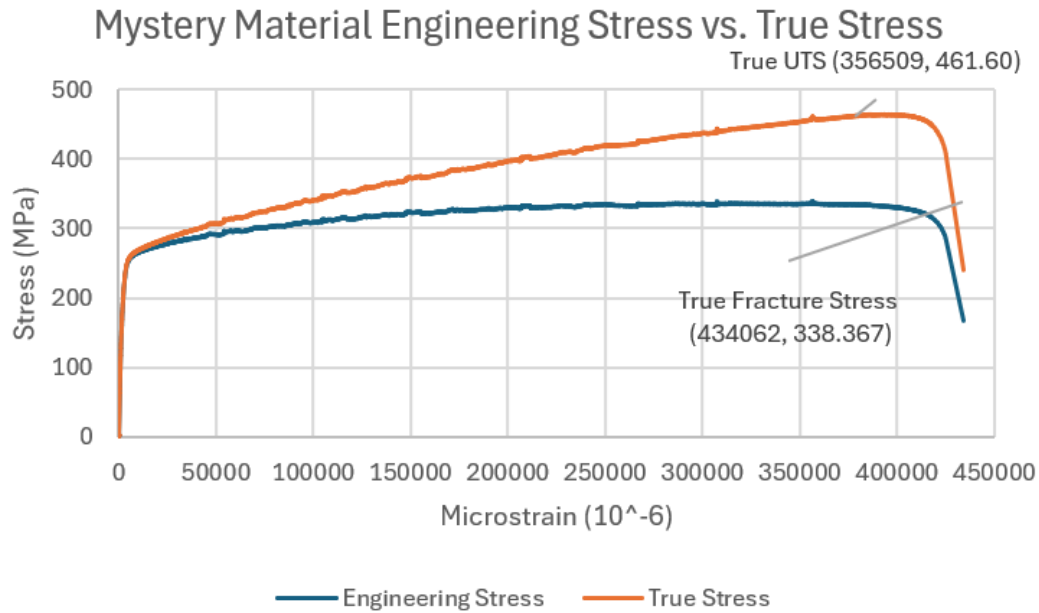


Figure 6. Engineering stress vs. true stress curves for the mystery specimen.

Table VI. Justification of the identity of the mystery material.

Material	Guess	Justification
Mystery Specimen	260 Brass	The mystery specimen is most likely 260 brass for four main reasons. First, the Young's Modulus of 260 brass is very similar to the mystery specimen at 16,000 ksi or 110.32 GPa. Second, both the specimen and 260 brass have considerably high ductility. This means that the material can withstand a large amount of plastic deformation before breaking. Thus, both the specimen and 260 brass experience considerable elongation before failure, as seen in the high true fracture strain in Table V. Third, 260 brass is much more likely to be the identity of the specimen compared to 353 brass. The main difference between 260 brass and 353 brass is that 353 brass contains a percentage of lead that significantly lowers the material's ductility. Since the mystery specimen had a considerably

		high tolerance to plastic deformation, it was possible to confidently identify its material. Finally, through personal experience playing a brass instrument, the mystery specimen was qualitatively very similar in both appearance and feel.
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References

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